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Density-Aware Graph Geodesic Stress Testing on a Learned Macroeconomic Manifold

by

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Abstract

[abstract]

Acknowledgments

[acknowledgments]

Own Work Declaration

[own work declaration]

AI disclosure

[see guidance document]

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1 Introduction

James thinks the intro might have too many sections. Maybe not all necessary.

1.1 motivation

Limitations of linear interpolation through Euclidean space; it gives unrealistic combinations of variables.

Each month gives many observations, but these could be represented using far fewer dimensions.

1.2 Research problem

- learn low-dimensional representation of macro-financial states;
- construct stress transitions that respect the learned geometry and historical density;
- determine whether temporal regimes emerge in this latent representation.

1.3 Research questions

Main questions:

- Can diffusion maps recover a stable and interpretable low-dimensional representation of macro-financial conditions?
- Do graph-based and density-aware stress paths remain in more historically plausible regions than direct linear interpolation?

Extension questions:

- Do hidden Markov regimes fitted in diffusion space correspond to recognised economic regimes, including NBER recessions?
- Do density-aware geodesics pass through the inferred regimes in a temporally and economically coherent way?

1.4 contributions

Have done:

- Implementation and numerical validation of a diffusion-map pipeline;
- Construction of density-aware graph geodesics for macroeconomic stress testing;
- Lifting of latent stress paths into economically interpretable variables

Aim to do:

- Fitting of a Gaussian HMM to diffusion coordinates rather than the original high-dimensional observations;
- Joint evaluation of geometric paths, historical density, and temporal regimes.

1.5 Link to JDKF paper??

Should I explain how this dissertation was motivated by the JDKF paper, with the observation process driven by a lower-dimensional latent process? I will put this somewhere, but not sure if I should put it here.

2 Background

2.1 High-dimensional state vectors and the manifold hypothesis

A macroeconomic state vector is a point

$$z_t = (z_t^1, \dots, z_t^D) \in \mathbb{R}^D$$

representing a variety of different macroeconomic variables at a time t, e.g. fuel prices, interest rates, etc. Although D may be large, the collection of observed states may concentrate near a lower-dimensional set

$$\mathcal{M} \in \mathbb{R}^D.$$

This is the manifold hypothesis. In this dissertation, the manifold is not assumed known analytically; instead, it is estimated numerically from data.

2.2 Latent diffusions and spectral coordinates

Explanation of the latent process and how the diffusion map is suitable for this model. This is heavily based on the JDKF paper.

Assumed latent process:

$$d\theta_t = -\nabla U(\theta_t), dt + \sqrt{2} dW_t,$$

and its infinitesimal generator:

$$\mathscr{L} = \Delta - \nabla U \cdot \nabla.$$

Generator eigenfunctions provide dynamically meaningful coordinates. The essential relation is

$$\psi_j(z_t) \approx \varphi_j(\theta_t)$$

where ψ_j is obtained from the finite-data graph and φ_j is an eigenfunction of the limiting differential operator.

2.3 Diffusion maps

This outlines how we actually build the diffusion maps, rather than the theory above.

Diffusion maps are a nonlinear dimensionality-reduction technique which use local similarities between observations to recover the geometry of a data set [2]. Suppose that

$$\mathcal{Z} = \{z_1, \dots, z_N\} \subset \mathbb{R}^D$$

is a collection of observations sampled from, or concentrated near, a lower-dimensional manifold

$$\mathcal{M} \subset \mathbb{R}^D.$$

The method constructs a weighted graph on the observations and uses the spectral properties of a random walk on this graph to obtain intrinsic coordinates for the data.

The first step is to calculate the pairwise squared Euclidean distances

$$\delta_{ij} = \|z_i - z_j\|^2.$$

These distances are converted into affinities using the Gaussian kernel

$$W_{ij} = \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right), \tag{2.1}$$

where the bandwidth $\varepsilon > 0$ controls the scale over which two observations are regarded as neighbours. The matrix

$$W = (W_{ij})_{i,j=1}^N$$

is symmetric and contains values between zero and one. Nearby observations have affinity close to one, whereas the affinity between distant observations is close to zero. Since each observation has zero distance from itself, $W_{ii} = 1$.

The weighted degree of observation i is defined by

$$q_i = \sum_{j=1}^{N} W_{ij}. \tag{2.2}$$

Unlike the degree of an unweighted graph, q_i does not simply count the number of adjacent vertices. It measures the total affinity of z_i to the sample. Consequently, a large value of q_i indicates that z_i has many nearby observations, or several observations to which it has strong affinity. Conversely, a value close to one indicates that almost all of the degree is contributed by the diagonal entry $W_{ii} = 1$, and hence that z_i is nearly isolated at the chosen bandwidth. The degree therefore acts as a kernel-based measure of local sampling density.

If the observations are sampled non-uniformly from $\mathcal{M}$, densely sampled regions may dominate the resulting random walk. Diffusion maps allow this effect to be controlled through the density-normalised affinity matrix

$$W_{ij}^{(\alpha)} = \frac{W_{ij}}{q_i^{\alpha} q_j^{\alpha}}, \qquad \alpha \in [0, 1]. \tag{2.3}$$

The parameter α determines how strongly the empirical sampling density is removed. The choice $\alpha = 0$ leaves the original affinities unchanged, whereas $\alpha = 1$ removes the leading influence of non-uniform sampling density and, under standard asymptotic conditions, yields an approximation to the Laplace–Beltrami operator associated with the underlying manifold [2]. Intermediate choices retain some dependence on the sampling distribution.

A second degree vector is then formed from the normalised affinities:

$$d_i^{(\alpha)} = \sum_{j=1}^{N} W_{ij}^{(\alpha)}. \tag{2.4}$$

Writing

$$D_\alpha = \operatorname{diag}\left(d_1^{(\alpha)}, \ldots, d_N^{(\alpha)}\right),$$

the diffusion transition matrix is

$$P = D_\alpha^{-1} W^{(\alpha)}, \qquad P_{ij} = \frac{W_{ij}^{(\alpha)}}{\sum_{k=1}^{N} W_{ik}^{(\alpha)}}. \tag{2.5}$$

Each row of P sums to one, so P can be interpreted as the transition matrix of a Markov chain on the data. In particular, P_{ij} is the probability of moving from observation z_i to observation z_j in one step. The entries of P^t give the corresponding transition probabilities after t steps.

This random-walk interpretation induces a notion of distance that depends on the connectivity of the data rather than solely on the ambient Euclidean distance. Two observations are considered close if random walks beginning at them have similar transition probabilities after t steps. The resulting diffusion distance may be written as

$$D_t^2(z_i, z_j) = \sum_{k=1}^{N} \frac{\left(P_{ik}^t - P_{jk}^t\right)^2}{\pi_k}, \tag{2.6}$$

where π_k denotes the stationary probability associated with observation z_k. This compares the full distributions of possible destinations from the two starting points. Consequently, diffusion distance is robust to individual short-circuit connections and reflects the presence of multiple paths through the data cloud.

Although P is generally not symmetric, it is similar to the symmetric matrix

$$S = D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}. \tag{2.7}$$

Indeed,

$$P = D_\alpha^{-1/2} S D_\alpha^{1/2}.$$

The matrices P and S therefore have the same eigenvalues. Since S is symmetric, its eigenvectors are orthogonal and its eigendecomposition is numerically more stable than directly diagonalising P. If

$$S u_k = \lambda_k u_k,$$

then the corresponding right eigenvector of P is

$$\phi_k = D_\alpha^{-1/2} u_k.$$

The eigenvalues are ordered as

$$1 = \lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \cdots,$$

where the first eigenvector ϕ_0 is constant and carries no geometric information. The diffusion-map embedding at diffusion time t is therefore defined using the leading non-trivial eigenpairs:

$$\Psi_t(z_i) = \left(\lambda_1^t \phi_1(i), \ldots, \lambda_r^t \phi_r(i)\right) \in \mathbb{R}^r. \tag{2.8}$$

The eigenvalue factors suppress rapidly decaying modes, while the eigenvectors describe the dominant directions of diffusion through the data. With the complete eigendecomposition, Euclidean distance in diffusion-coordinate space is equal to diffusion distance:

$$D_t^2(z_i, z_j) = \sum_{k \geq 1} \lambda_k^{2t} \left(\phi_k(i) - \phi_k(j)\right)^2.$$

Truncating this expansion after r terms provides a low-dimensional approximation that retains the most persistent diffusion modes.

Under suitable assumptions on the sampling process, together with an appropriate relationship between the sample size and the bandwidth, the graph-based operator converges to a differential operator on $\mathcal{M}$. Correspondingly, its eigenvectors converge to evaluations of the eigenfunctions of the limiting operator at the sampled observations. This provides the theoretical basis for interpreting diffusion coordinates as empirical approximations to intrinsic spectral coordinates of the underlying system. In the latent-dynamical setting considered in Section 2.2, this connection motivates the use of diffusion maps to obtain data-driven coordinates for an otherwise unobserved state space.

The bandwidth ε is central to the construction. If ε is too small, the off-diagonal affinities become negligible, so that

$$W \approx I$$

and many observations are effectively isolated. In this regime, the diffusion operator describes a poorly connected collection of local components. If ε is too large, most affinities approach one, so that

$$W \approx \mathbf{1}\mathbf{1}^\top,$$

and the kernel loses its ability to distinguish local geometric structure. An appropriate bandwidth must therefore be sufficiently large to connect the observations while remaining sufficiently local to preserve the geometry of the data.

One diagnostic for the bandwidth is the kernel mass

$$T(\varepsilon) = \sum_{i,j=1}^{N} \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right). \tag{2.9}$$

For observations sampled from a sufficiently regular d-dimensional manifold, the kernel mass is expected to satisfy

$$T(\varepsilon) \approx C\varepsilon^{d/2}$$

over an appropriate range of local scales. It follows that

$$\frac{d \log T(\varepsilon)}{d \log \varepsilon} \approx \frac{d}{2}. \tag{2.10}$$

A stable region in the log–log slope can therefore provide information about both the intrinsic dimension and a suitable range of bandwidths. In finite, noisy, and non-uniformly sampled data, however, such a scaling region may be weak or absent. Kernel-mass scaling is therefore best treated as a diagnostic and considered alongside graph connectivity, the degree distribution, spectral stability and the quality of the resulting coordinates.

The embedding dimension r determines how many non-trivial diffusion modes are retained. A common diagnostic is the presence of a spectral gap. If

$$\lambda_r - \lambda_{r+1}$$

is large relative to neighbouring gaps, the first r non-trivial eigenvectors represent comparatively persistent modes, whereas subsequent eigenvectors decay more rapidly and may primarily reflect fine-scale variation or noise. Spectral gaps do not provide an infallible choice of dimension, particularly when eigenvalues occur in near-degenerate groups. The final dimension should therefore also be assessed through stability under nearby bandwidths, preservation of relevant geometric structure, and the requirements of the downstream analysis.

2.4 Kernel scale and spectral diagnostics

Explanation of the theory behind choosing our model parameters basically. The kernel mass plots, spectral analysis, and the justification of the embedding dimension. Also graph connectivity and robustness of all choices.

- kernel mass $T(\varepsilon)$;
- the scaling relation $T(\varepsilon) \propto \varepsilon^{d/2}$;
- spectral gaps;
- eigenvalue multiplicities;
- graph connectivity;
- stability across nearby parameter values.

2.5 Graph geometry and shortest paths

Explain how we construct our graph on the diffusion coordinates.

- Graph paths;
- ordinary graph-geodesic cost;
- the distinction between a Euclidean chord and a path constrained to observed neighbourhoods;
- Dijkstra's shortest-path problem.

2.6 Density-aware graph geometry??

This is what we are adding to the literature, so is it worth going into extra detail here? Or just the same detail as above?

2.7 Hidden Markov models??

This is the extension, and I am not sure if it will be done

2.8 Positioning within the literature??

Should I have this as its own section or do I continually reference the literature throughout this section?

3 Data and Preprocessing

3.1 Data sources and sample period

3.2 Variable selection

3.3 Temporal alignment and missing observations

3.4 Transformations

3.5 Data diagnostics

3.6 Final analysis matrix (and sample splits, if HMM)??

4 Numerical methods

4.1 Overview of the computational pipeline

This is where you give a clear picture of what you actually are going to do. Reference the different sections, etc.

raw data $\to$ transformation $\to$ diffusion map $\to$ density and graph $\to$ stress paths $\to$ lifting

? $\to$ HMM regime analysis

In this section, we describe the entire computational pipeline. Firstly, using our transformed and standardised data, we find a diffusion map, which captures the potential non-linear structure of the macroeconomic manifold on which the data lies. Using that diffusion map, we choose an appropriate embedding dimension, and now we have our new coordinate system. Using these new coordinates, we construct a k-nearest-neighbours graph and compute an estimation of the densities. We then implement our density-aware graph path model to interpolate a path between a variety of specifically chosen economic endpoints. We compare this path to various other types of paths, specifically, a linear path in diffusion space (Is this also a linear path in macro space?), and a standard graph path. Then we can lift these paths back to the macro space, and there we can compare different paths in real coordinates to see how well our model does.

4.2 Diffusion-map implementation - Example

The diffusion-map construction described in Section 2.3 was applied to the transformed macroeconomic state matrix

$$\widetilde{Z} = \begin{pmatrix} \widetilde{z}_1^\top \\ \vdots \\ \widetilde{z}_N^\top \end{pmatrix} \in \mathbb{R}^{N\times D},$$

where each row represents one monthly observation and each column corresponds to a macroeconomic or financial variable. The transformation and construction of this matrix are described in Section 3. The NBER recession indicator was retained for subsequent validation but was not included as an input to the embedding.

Pairwise squared Euclidean distances were first calculated between all transformed observations. These distances were converted into Gaussian affinities using the diffusion bandwidth ε. The density-normalisation parameter was fixed at $\alpha = 1$. This choice reduces the influence of variations in the empirical sampling density, so that the resulting operator is intended to reflect the geometry of the underlying state space rather than simply assigning greater importance to densely sampled periods. This is my understanding, I will need a citation here I imagine?

For numerical stability, the eigenpairs were obtained from the symmetric matrix associated with the row-stochastic diffusion operator, rather than by directly diagonalising the generally non-symmetric transition matrix. The eigenvalues were arranged in descending order and the corresponding right eigenvectors of the transition matrix were recovered from the symmetric eigenvectors. The first eigenvector, associated with the trivial eigenvalue $\lambda_0 = 1$, was omitted. At diffusion time $t = 1$, the r-dimensional coordinate assigned to observation i was

$$\Psi(\widetilde{z}_i) = \big(\lambda_1\phi_1(i), \ldots, \lambda_r\phi_r(i)\big).$$

Thus, the embedding retained both the leading non-trivial eigenvectors and their associated eigenvalue scaling.

The bandwidth ε controls the locality of the Gaussian kernel citation? and was therefore selected before fixing the embedding dimension. As an initial diagnostic, the kernel mass

$$T(\varepsilon) = \sum_{i,j} \exp\left(-\frac{\|\widetilde{z}_i - \widetilde{z}_j\|^2}{\varepsilon}\right)$$

was evaluated over a logarithmically spaced range of bandwidths. Its local log–log slope was computed numerically, as described in Section 2.4. In the empirical data, however, selecting the bandwidth from the maximum local slope produced a highly localised kernel containing observations with degree close to one. Since the diagonal affinity satisfies $W_{ii} = 1$, a degree near one indicates that an observation has negligible affinity with the remainder of the sample. The kernel-mass rule was therefore treated as a preliminary diagnostic rather than as an automatic selection criterion.

The final bandwidth was selected by considering three complementary properties of the resulting diffusion operator. First, the minimum kernel degree was required to be sufficiently above one, avoiding observations that were effectively isolated. Second, the coefficient of variation of the kernel degrees was examined to assess the stability of the connectivity structure across nearby bandwidths. Third, the leading eigenvalues and the gaps between consecutive eigenvalues were compared to determine whether the dominant spectral structure remained stable under small changes in ε. The behaviour of the individual diffusion coordinates was also inspected to exclude bandwidths for which the leading coordinates were almost constant apart from a small number of isolated spikes.

On the basis of these diagnostics, the bandwidth was fixed at

$$\varepsilon = 3. \quad \text{Do I put this here? Or do I put it once I show the results of the tests?}$$

At this value, the minimum kernel degree was approximately 19.35, compared with values close to one at smaller bandwidths, while the leading eigenspectrum and diffusion coordinates were stable relative to neighbouring choices. The detailed degree statistics and spectral comparisons are reported in Section 6.1.

The embedding dimension was selected using the leading non-trivial eigenspectrum together with the stability and empirical interpretability of the resulting coordinates. Additionally, investigation of near-collision pairs in various dimensions revealed how much information was added at each dimension. [I initially thought that this choice was simple, but when trying to justify it, it became trickier. There are a few tests, and I am still unsure about the conclusion. A slight discussion is likely to happen around this choice, so I am not sure if I should explicitly state the dimension choice of $r = 3$ here. Similar questions about ε, but the choice of dimension is even less trivial]. A three-dimensional representation was retained,

$$r = 3,$$

as the first three non-trivial coordinates preserved the dominant spectral structure and produced clearer separation of economically distinct observations than the corresponding two-dimensional projection. This was assessed by visualising the embedding after colouring observations by calendar time, recession status, unemployment, inflation, interest rates, corporate bond yields and industrial production. These visualisations are presented and discussed in Section 6.1.

The final diffusion-map configuration used throughout the subsequent graph, stress-path and hidden-regime analyses was therefore

$$\alpha = 1, \qquad \varepsilon = 3, \qquad r = 3.$$

Robustness to nearby bandwidths and alternative embedding dimensions is considered in Section 6.3.

Algorithm 1 Diffusion map - would it be helpful to add this?

1: **Input:** $\tilde{Z} \in \mathbb{R}^{N\times D}$, ε, α, r
2: **Output:** Embedding coordinates $\Psi \in \mathbb{R}^{N\times r}$
3: $W_{ij} \leftarrow \exp(-\frac{\|z_i - z_j\|^2}{\varepsilon})$
4: $q_i \leftarrow \sum_{j=1}^{N} W_{ij}$
5: $W_{ij}^{(\alpha)} \leftarrow \frac{W_{ij}}{q_i^\alpha q_j^\alpha}$
6: $d_i^{(\alpha)} \leftarrow \sum_{j=1}^{N} W_{ij}^{(\alpha)}$
7: $D_\alpha \leftarrow \mathrm{diag}\left(d_1^{(\alpha)}, \ldots, d_N^{(\alpha)}\right)$
8: $P \leftarrow D_\alpha^{-1} W^{(\alpha)}$
9: $S \leftarrow D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}$
10: Compute the eigendecomposition $S\phi_k = \lambda_k \phi_k$.
11: Sort the eigenpairs in descending order of λ_k.
12: Discard the trivial eigenpair (λ_0, ϕ_0).
13: Form $\Psi \leftarrow [\lambda_1\phi_1, \ldots, \lambda_r\phi_r]$.
14: **return** Ψ

4.3 Density estimation and graph construction & Stress-path constructions??

To create scenario paths between two economic points in diffusion space ψ_{start} to ψ_{stress}, we build a k-nearest-neighbours graph and use SciPy's shortest_path function to find a path

$$\Gamma = (\psi_{i_1}, \ldots, \psi_{i_\ell}),$$

where $\psi_{i_1} = \psi_{\text{start}}$ and $\psi_{i_\ell} = \psi_{\text{stress}}$. The shortest_path algorithm solves the following minimisation problem

$$\Gamma = \underset{\Gamma: i_{\text{start}} \to i_{\text{end}}}{\arg\min} \sum_{(i,j)\in\Gamma} w_{ij},$$

where w_{ij} is the edge cost of our graph. This is usually defined as $w_{ij} = \|\psi_i - \psi_j\|$ citation?.

With the diffusion coordinates from Section 4.2 we build a k-nearest-neighbours graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \quad \mathcal{V} = \{1, \ldots, N\}.$$

In our implementation, we used Scikit-Learn's NearestNeighbours class to create graphs of different values of k. We then symmetrise the graphs by taking unions of directed nearest-neighbour edges. This results in each node having slightly more than k neighbours on average. Checks are performed on each resulting graph to ensure that the graph is connected; the graph is not nearly complete; the typical node degree is reasonable; a few artificial jumps do not dominate shortest paths; and the results are stable for nearby values of k.

Next, we find the estimated density of points in our diffusion embedding by the method discussed in Section 2.6. Namely, we calculate

$$\hat{\rho}_i = \sum_{j=1}^{N} \exp\left(-\frac{\|\psi_i - \psi_j\|^2}{h}\right),$$

where h is a bandwidth parameter, chosen to provide a suitable variation in density values. We store the mean of the density vector

$$\bar{\rho} = \frac{1}{N}\sum_{i=1}^{N} \hat{\rho}_i$$

as a scale value, which is used to normalise $\hat{\rho}$ as follows:

$$\rho_i = \frac{\hat{\rho}_i}{\bar{\rho}}.$$

We also store $\hat{\rho}$ in order to calculate the density of the linear path in diffusion space, described in Section 4.4. For computational stability, we define a density floor $\delta \ll 1$ and impose that $\rho_{i,\delta} = \max\{\rho_i, \delta\}$. Now we can find our density potential defined as

$$V_i = -\log(\rho_{i,\delta}).$$

We can now define our density-aware graph cost, for $\beta \geq 0$,

$$w_{ij}^{(\beta)} = \|\psi_i - \psi_j\| \exp\left(\beta \frac{V_i + V_j}{2}\right).$$

We see that for $\beta = 0$ corresponds to the standard graph path (/ ordinary graph geodesic), whereas for $\beta > 0$, traveling through low-density regions become more expensive. The density-aware graph paths are simply calculated by passing $w_{ij}^{(\beta)}$ into the shortest_path function to return Γ_β.

We run tests on many values of β and visually inspect the resulting paths to choose an optimal value.

The naive path between the two points

$$\gamma(s) = (1-s)\psi_{\text{start}} + s\psi_{\text{end}}$$

is simply constructed with NumPy's 'linspace' function where we choose a suitable number of grid points.

Should I add the part where I test whether density corresponds to plausibility? Should there be something about this in the background?

4.4 Stress-path evaluation

We analyse the paths in diffusion space by calculating the density of the points along the paths. Our main result metric is the minimum density along a given path. For the graph paths, each point already corresponds to a density ρ_i; however, the linear paths $\gamma(s)$ cut through the diffusion space, so we need to compute the density of points along the interpolated line. We define the density along a discretised linear path in diffusion space as

$$\hat{\rho}(\gamma_i) = \sum_{j=1}^{N} \exp\left(-\frac{\|\gamma_i - \psi_j\|^2}{h}\right),$$

where h is the same bandwidth used to calculate the density of the diffusion coordinates in Section 4.3. We then normalise this density vector $\hat{\rho}(\gamma)$ by dividing each element by the mean density of all diffusion coordinates

$$\rho(\gamma) = \frac{\hat{\rho}(\gamma)}{\bar{\rho}},$$

which ensures that the density of the linear path is directly comparable with the density of the diffusion coordinates. For numerical stability, we then apply the same density floor as we do for the diffusion coordinate densities by taking $\rho(\gamma_i) = \max(\rho(\gamma_i), \delta)$. We can then also find the density potential for the points on the linear path by taking $V(\gamma_i) = -\log(\rho(\gamma_i))$. The primary metric is the minimum density along the path:

$$M_{\min}(\gamma) = \min_s \rho(\gamma(s)).$$

For graph paths, this is

$$M_{\min}(\Gamma) = \min_{i\in\Gamma} \rho_i,$$

and we aim to show that stress paths avoid historically implausible low-density regions. We also calculate the following secondary metrics:

Mean path density.

$$M_{\text{mean}}(\Gamma) = \frac{1}{L+1}\sum_{\ell=0}^{L} \rho_{i_\ell}$$

Latent path length.

$$L_{\text{latent}}(\Gamma) = \sum_{\ell=0}^{L-1} \|\psi_{i_{\ell+1}} - \psi_{i_\ell}\|$$

Density-weighted cost.

$$C_\beta(\Gamma) = \sum_{(i,j)\in\Gamma} \|\psi_i - \psi_j\| \exp\left(\beta \frac{V_i + V_j}{2}\right)$$

Smoothness per unit length. For a discretised path $\gamma_0, \dots, \gamma_L$, define:

$$S(\gamma) = \sum_{\ell=1}^{L-1} \|\gamma_{\ell+1} - 2\gamma_\ell + \gamma_{\ell-1}\|^2 / L(\gamma), \quad L(\gamma) = \sum_{i=0}^{L-1} \|\gamma_{i+1} - \gamma_i\|$$

Direction change

$$D(\gamma) = \sum_{\ell=0}^{L-1} \|u_{\ell+1} - u_\ell\|^2, \quad u_\ell = \frac{d_\ell}{\|d_\ell\|}, \quad d_\ell = \gamma_{\ell+1} - \gamma_\ell$$

Ambient plausibility. For lifted macroeconomic paths:

$$D_{\text{NN,max}} = \max_\ell d_{\text{NN}}(z_\ell), \quad D_{\text{NN,mean}} = \frac{1}{L+1}\sum_{\ell=0}^{L} d_{\text{NN}}(z_\ell)$$

We also compute the distance from each path to its nearest neighbour in diffusion space; this gives us a notion of how plausible the specific point is. If the point is very far from an observed point in diffusion space, then it may not be a realistic point; however, if a point is very near an observed point, then it must be plausible. I kind of just made this up; I think I need to justify why we want our points to have a higher density; the diffusion mapping density doesn't necessarily correlate to how likley/plausable a point is, only that it is near other points in diffusion space. By construction, for any graph path, the distance to the nearest neighbour in diffusion space is zero, because the graph paths jump between observed nodes. To find the distances from the linear path γ, a nearest-neighbour search structure was fitted to the historical diffusion coordinates using SciPy's NearestNeighbour class. For each interpolated path point γ_i, the diagnostic records its Euclidean distance to the closest observed state,

$$d_{NN}(\gamma_i) = \min_{1\le j\le N} \|\gamma_i - \psi_j\|.$$

Small values indicate that the path remains close to historically observed states, whereas large values indicate movement away from the empirical point cloud.

4.5 Lifting paths into macroeconomic space & Path-evaluation criteria

In this dissertation, lifting graph paths to macroeconomic space is simply a matter of matching indices, because each diffusion coordinate is associated with each macroeconomic variable, and graph paths strictly follow diffusion coordinates. To lift the linearly interpolated lines into macroeconomic space requires more work. We compare two different options: a linear lift $\hat{z}(s) = H(\gamma(s) - \bar{\psi}) + \bar{z}$, where $\hat{\psi}$ is the mean of the diffusion coordinates across time,

$$H = \underset{B \in \mathbb{R}^{D \times r}}{\arg\min} \sum_{i=1}^{N} \|\tilde{z}_i^{(c)} - B\psi_i^{(c)}\|^2,$$

where $\tilde{z}_i^{(c)}$ and $\psi_i^{(c)}$ are the coordinates centred by subtracting their respective means across time. We need to ensure the values $\tilde{z}^{(c)}$ are centred because the lifting operator H cannot learn an intercept. This method is used by Baker et. al [1] and thus we call it the 'Bakery-style lift'. We also experiment with a local-neighbour lift (LNL) approach

$$\hat{z}(s) = \sum_{j \in \mathcal{N}_m(\gamma(s))} a_j(s)\tilde{z}_j,$$

where $\mathcal{N}_m(\gamma(s))$ are the m nearest training points to $\gamma(s)$ in diffusion space and the weights satisfy

$$a_j(s) \geq 0, \quad \sum_j a_j(s) = 1.$$

In this dissertation, we use Gaussian weights

$$a_j(s) = \frac{\exp(-\|\gamma(s) - \psi_j\|^2/\tau)}{\sum_{j \in \mathcal{N}_m(\gamma(s))} \exp(-\|\gamma(s) - \psi_k\|^2/\tau)}.$$

A simple grid search is used to find the optimal value of m for our training set. And for both methods, when interpolating between two known economic endpoints, we enforce each endpoint being the exact lifted macroeconomic coordinate rather than the given coordinate from the method.

We then investigate the reconstruction diagnostics. We calculate a leave-one-out RMSE and R^2 statistics for each lifting method and compare them to a baseline lifting method where we simply lift each point to the mean position of the training data points. The statistics are computed by looping through each training point and fitting each lifting method to the rest of the points; the held-out diffusion coordinate is then lifted into the data cloud. We then store the lifted coordinates of each training point and compare them to the real point. We can compute the RMSE and the R^2 values for each method.

Next, we investigated the plausibility diagnostics, where we find how close each lifted point on a linear path is to its nearest observed training point. We compare the Baker-style method and the LNL method for each pair of dates from benign to crisis. For each point along a path, we calculate the distance to its nearest neighbour in macroeconomic space, similar to in Section 4.4 we define

$$D_{NN}(\hat{z}(s)) = \min_{1 \leq j \leq N} \|\hat{z}(s) - \tilde{z}_j\|.$$

We compare these values to the closest nearest neighbours of the entire training set. Specifically, we evaluate the median values of D_{NN} and the 95% percentile value of D_{NN} for each path and method. We compare this to the same statistics for the training set nearest neighbours vector.

4.6 Nyström extension for out-of-sample lifting

In this subsection, we outline the method we use to embed out-of-sample macroeconomic coordinates into our diffusion coordinates without re-running the entire eigendecomposition described in Section 2.3. This method requires us to use the eigenpairs $\{(\lambda_\ell, \phi_\ell)\}$, the degree vector q_i, and the bandwidth parameter ε from Algorithm 1 trained on our original dataset $\tilde{Z}$. Suppose we have a coordinate $\tilde{z}_{\text{new}} \in \mathbb{R}^D$ in macroeconomic space and we want to embed it into $\mathbb{R}^r$. We first compute an affinity vector

$$w_{\text{new},i} = \exp\left(-\frac{\|z_{\text{new}} - z_i\|^2}{\varepsilon}\right)$$

and the corresponding degree for this point

$$q_{\text{new}} = \sum_{i=1}^{N} w_{\text{new},i}.$$

We then apply the α-normalisation, which removes the effect of the sampling distribution (as described in Section 2.3)

$$w_{\text{new},i}^{(\alpha)} = \frac{w_{\text{new},i}}{q_{\text{new}}^{\alpha} q_i^{\alpha}},$$

and compute the row-normalised vector

$$\tilde{w}_{\text{new},i} = w_{\text{new},i}^{(\alpha)} \sum_{j=1}^{N} w_{\text{new},i}^{(\alpha)}.$$

The Nyström extension then extends the eigenfunctions ϕ_ℓ to the new points via

$$\phi_\ell(z_{\text{new}}) = \frac{1}{\lambda_\ell} \sum_{i=1}^{N} \tilde{w}_{\text{new},i} \phi_\ell(i),$$

and the extended diffusion coordinates are therefore

$$\Psi(z_{\text{new}}) = (\lambda_1 \phi(z_{\text{new}}), \ldots, \lambda_r \phi(z_{\text{new}})).$$

This is the exact same method used in [3]; however, we add the α-normalisation step.

4.7 Gaussian HMM specification??

In addition to checking the density in diffusion space and inspecting the macroeconomic variables along paths, we also wish to examine our paths within a Gaussian Hidden Markov Model (HMM).

To choose our HMM specification, i.e. the number of states K, we fit an HMM to our diffusion coordinates for $K \in \{2, 3, 4, 5, 6, 7\}$. For each choice of K, we fit 100 models with different random prior initialisations and store the model with the best log-likelihood score, $(\log L)_K$. Definitely need to give some more context here. For each model, we can calculate the total number of parameters used via the following formula:

$$p = (K-1) + K(K-1) + Kr + K\frac{r(r+1)}{2},$$

where r is the dimension of our diffusion embedding. We can calculate the Bayesian Information Criterion (BIC) score

$$\text{BIC}_K = p \log N - 2(\log L)_K,$$

where N is the number of datapoints used to fit the HMM. The BIC score gives us a score balancing the complexity of the model and the model fit. We seek the model with the lowest BIC score. We can see that increasing the model complexity (increasing K) increases the BIC score, so the model must then improve its log-likelihood sufficiently to compensate.

For each value of K, we also compute an out-of-sample log-likelihood score. We fit the HMM with the first 70% of diffusion coordinates, and we evaluate how well this model's parameters explain the remaining 30% of coordinates, which have been Nyström lifted into the diffusion space (see Section 4.6).

The choice of K is not exactly trivial here because $K = 6$ wins on BIC, and $K = 7$ wins of oos log-likelihood. Should I just choose a K value here and mention these tests or should I include this part in the results section where I discuss why I choose my K value?

After choosing our model, we validate our choice by investigating the real values of the macroeconomic variables in each predicted state. The HMM does not have any notion of what each state is; it just identifies that different points in the time series seem to come from different distributions. We investigate the means of each different state and try to infer an economic interpretation.

We then use the NBER recession indicators to further validate the model; if one of the HMM's predicted states lines up with the NBER recession dates, then we can infer that that state is a recession state, giving credence to the model. Accordingly, Cohen's κ was calculated to compare each state to the ground truth NBER indicator variable. Cohen's κ value gives a measure of how well the HMM predicts the NBER state, adjusting the score to account for random chance. It is defined as

$$\kappa = \frac{p_0 - p_e}{1 - p_e},$$

where p_o is the proportion of time where the state lines up with the NBER state, and p_e is the proportion of the time where the states are expected to match due to random chance. For example, if 90% of the dates are marked as non-NBER recessions, and the HMM model randomly assigned 80% of the timepoints to 0 for a certain state, then the random chance of the HMM matching the NBER states is $(0.9 \cdot 0.8) + (0.1 \cdot 0.2) = 0.74$. The model would need to then match with the NBER states more than 74% of the time to have a positive Cohen's κ score.

Finally, we investigate the embedding density level grouped by the HMM state predictions. We do this because we know that low density corresponds to economic outliers, therefore we would expect there to be a clear difference in the mean density levels for nodes in each state. (at this point, we wouldn't have shown the results for densty $\propto$ historical support, so can I say this? Maybe include those results above???

Once we justify the use of the HMM model, we can say that it can appropriately model our system. We can then ask the question "how likley is it that this path is explained by the fitted model?" for each of our different paths. We generate stress paths using linear interpolation through diffusion space, ordinary graph cost and density-aware graph cost by the method explained in Section 4.3. We then compute the log-likelihood of each path being explained by our fitted model parameters.

4.8 Robustness design

In this section, we describe the robustness checks we performed to ensure the validity of our results. We hope to see that our qualitative result is stable: density-aware graph paths avoid low-density regions more effectively than straight-line interpolation.

4.8.1 Varying parameters

Vary bandwidth in diffusion mapping. Here we can look at more results from the diffusion map choice at the start. We vary ε, and we look at the mean degree and the median edge length of the kNN graph constructed from the embedding. We also investigate the resulting paths from the new diffusion map; specifically, we monitor how the minimum density, the density-weighted cost, the number of points and the total length of each path change as we change the bandwidth.

Next we vary the number of nearest neighbours used in the construction of the kNN graph from the diffusion embedding (with the original bandwidth $\varepsilon = 3$). We then compute the same metrics as in the bandwidth experiment, with the addition of the direction-change score for each path.

Next, we simply investigate the effect of varying the penalty term β on the minimum density of the paths. This could be useful in deciding the strength of the density-seeking behaviour of our paths.

4.8.2 Varying density calculation

Now we compare different methods of calculating the density of the diffuion coordinates. We look at two common methods for computing the density of the diffusion coordinates and the points along the linear path, hence, we use x to represent a general point in the diffusion coordinate space. We still fit each density model to the diffusion coordinates, so distances to ψ are all gathered.

Alternative kernel density estimators

For a point $x \in \mathbb{R}^r$, the kernel density estimate is

$$\hat{\rho}_K(x) = \frac{1}{Nh^r} \sum_{j=1}^{N} K\left(\frac{\|x - \psi_j\|}{h}\right),$$

where $h > 0$ is the bandwidth and K is a radial kernel. Robustness is assessed using the following kernel choices.

Gaussian kernel

$$K_G(u) = \frac{1}{(2\pi)^{r/2}} \exp\left(-\frac{u^2}{2}\right).$$

Epanechnikov kernel

$$K_E(u) \propto \max\{1 - u^2, 0\}.$$

Exponential kernel

$$K_X(u) \propto e^{-u}.$$

To directly compare each method, we normalise each density estimate; therefore, the proportionality constants disappear. The Gaussian and exponential kernels have infinite support, while the Epanechnikov kernel has compact support. The exponential kernel decays more slowly with distance than the Gaussian kernel. I have inferred these support properties by looking at the graphs on the Scikit Learn documentation for the functions which compute these kernels. Do I cite that? We use Scikit Learn's KernelDensity class to compute each of these densities. Noticing that our degree density calculation contains $\exp\left(-\frac{\|x-\psi_j\|^2}{h}\right)$, and the Gaussian kernel KDE density contains $\exp\left(-\frac{\|x-\psi_j\|^2}{2h^2}\right)$ we can then set our bandwidth in the KDE function as $h_{\text{KDE}} = \sqrt{\frac{1}{2}h_{\text{dens}}}$. We should then see, after normalising the density estimates, the KDE with the Gaussian kernel match up with our graph-degree density exactly. We keep the same choice of bandwith for the Epanechnikov and exponential kernel to investigate the impact of changing the kernel only. For each kernel, the ordinary graph path remains the same; however, the density-aware graph path may change due to the effect of V in the cost function. We therefore recompute the density-aware graph path for each event and each density method. We then investigate whether or not the minimum density along paths increases from linear paths to density-aware paths.

4.8.3 Comparing lifted paths to PCA paths

Finally, as a linear baseline, principal component analysis (PCA) was applied to the quantile-standardised macroeconomic observations. The first r principal components were retained, giving a linear latent representation Ψ_{PCA}. For each event pair, a straight-line interpolation was constructed between the corresponding latent endpoints in PCA space. The interpolated path was then mapped back to macroeconomic space using the inverse PCA transformation. We employ the plausibility diagnostics from Section 4.5, namely we investigate the statistics of $D_{NN}(\hat{z}_{\text{PCA}}(s))$ and compare them to $D_{NN}(\hat{z}_{\text{LNL}}(s))$. We also plot a selection of lifted variables to inspect the paths visually.

5 Numerical Validation on Toy Model

5.1 Purpose of the validation experiments

5.2 Benchmark manifolds

5.3 Bandwidth and spectral validation

5.4 Recovery of manifold geometry

6 Macroeconomic Manifold and Stress Paths

This is the main results section of the paper

6.1 Empirical embedding results - example

6.2 Diffusion map of the macroeconomic data

The diffusion-map construction was applied to the quantile-transformed macroeconomic observations using the density normalisation $\alpha = 1$. The bandwidth and embedding dimension were selected using the diagnostics described in Section **??**. Since the purpose of this stage was to obtain a suitable geometric representation for the subsequent path analysis, rather than to interpret every individual diffusion coordinate, the results are reported only briefly.

Bandwidth selection. The kernel-mass scaling diagnostic suggested a bandwidth of approximately $\varepsilon = 1.13$. This value was rejected because the corresponding minimum kernel degree was close to one. As the diagonal affinity satisfies $W_{ii} = 1$, this indicated that some observations were almost isolated from the remainder of the sample. The leading diffusion coordinates at small bandwidths were also largely flat, with occasional sharp spikes, rather than displaying stable variation across the sample.

A bandwidth of

$$\varepsilon = 3$$

was selected instead. At this value, the minimum degree increased to approximately 19.35, and the degree distribution and leading eigenspectrum were stable relative to nearby bandwidths. Increasing the bandwidth further did not materially improve the leading coordinates and would have made the kernel progressively less local. The selected value was therefore the smallest bandwidth that provided a sufficiently connected and spectrally stable diffusion operator.

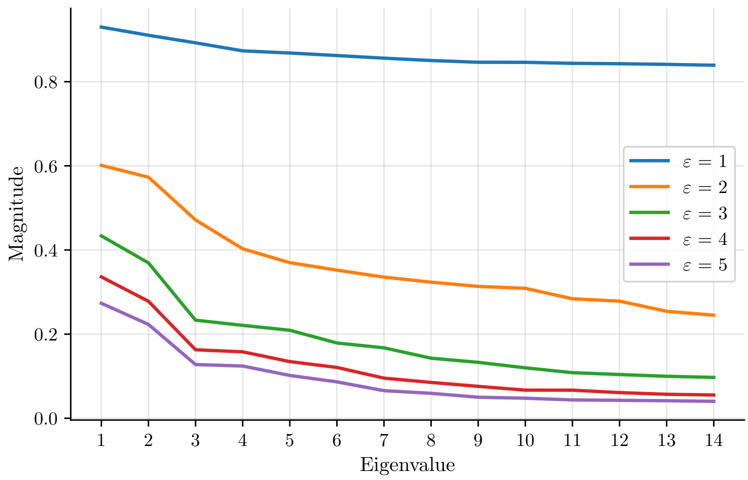


Figure 1: Spectrum of the transition matrix for different bandwidths.

ε	Min Degree	Degree CV
1	1.20	0.47
2	5.54	0.37
3	19.35	0.28
4	41.77	0.23
5	69.34	0.19

Table 1: Minimum degree and degree CV across bandwidths.

Embedding dimension. The leading eigenspectrum did not, by itself, provide an unambiguous cut-off for the embedding dimension. The incremental information supplied by additional coordinates was therefore assessed using the near-collision diagnostic. For each candidate dimension r, pairs of observations that were close in the first r diffusion coordinates (closest 1%) were identified, subject to a twelve-month temporal exclusion window. The separation of these pairs in the the higher-dimensional space was then measured and percentiles were taken.

Table 2 reports the results. The addition of coordinate $r = 3$ and $r = 4$ both produced meaningful separation of observations that appeared close in the lower-dimensional representation. By contrast, coordinate $r = 5$ provided substantially weaker separation, indicating that the incremental geometric information beyond dimension $r = 4$ was limited. The final embedding dimension was therefore fixed at

$$r = 4.$$

Table 2: Percent of the near-collision pairs that are no longer in the closest x% of pairs in the higher-dimensional embedding. Near-collisions are pairs are the top 1% of closest nodes in the lower-dimensional embedding after applying a twelve-month temporal exclusion window.

Comparison	Near-collision pairs	% Exit 1%	% Exit 5%	% Exit 10%
2D → 3D	2669	58.0	31.1	18.6
3D → 4D	2669	49.5	20.8	10.1
4D → 5D	2669	23.1	1.95	0.26

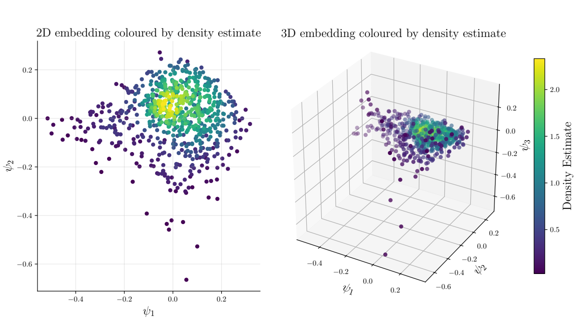
In Table 2, we can see that adding an extra dimension can add lots of separation. For example, roughly 496 of the near-collision pairs in $r = 2$ are no longer in the top 10% closest pairs in $r = 3$; this clearly shows that there is a considerable amount of new information gained by adding a 3rd coordinate. The results are similar for adding the 4th coordinate; however, there is a dramatic drop-off in the gained separation of near-collision nodes when adding the 5th coordinate.

Structure of the embedding. Figure 2 displays the resulting diffusion coordinates. Observations associated with NBER recessions were visibly displaced from the main concentration of observations in the leading projections. This separation was already present in the first few coordinates, indicating that recessionary periods represent comparatively unusual locations in the learned macroeconomic state space rather than being distinguished only by higher-order coordinates.

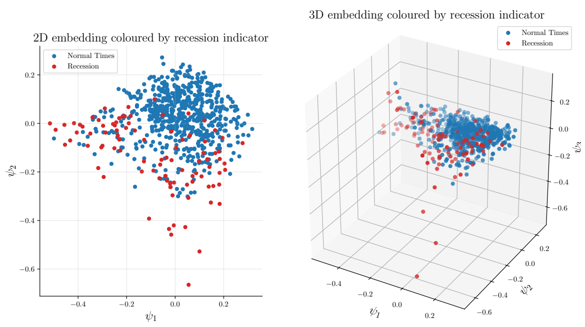
The embedding also exhibited systematic variation with NBER labels, inflation rates, and industrial production levels. These patterns support the interpretation of the diffusion coordinates as describing economically meaningful variation in the joint state of the variables. The coordinates themselves are not assigned direct economic labels, however, and are used primarily as a geometric state representation in the subsequent graph and stress-path analysis.

Overall, the selected diffusion map provided a connected and stable low-dimensional representation of the transformed macroeconomic data. This embedding is used as the state space for the graph-based stress paths constructed in the following section.

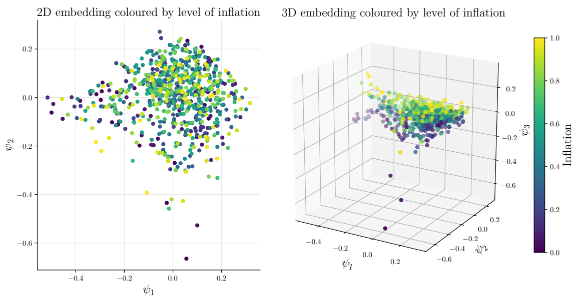
6.3 Robustness of parameters



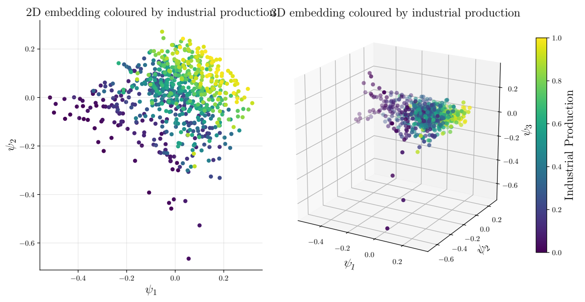
(a) Embedding Density



(b) Recession Periods



(c) Inflation



(d) Industrial Production

Figure 2: Diffusion-map representation of the quantile-transformed macroeconomic observations using $\alpha = 1$, $\varepsilon = 3$. While our diffusion dimension is 4-dimensional, here we only show 2D and 3D plots. Observations are coloured by density, NBER recession labels, inflation rates (quantile), and industrial production levels (quantile).

7 Hidden Regimes on the Macroeconomic Manifold

This is the extension

8 Conclusions and outlook

References

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[2] R. R. Coifman and S. Lafon. Diffusion maps. *Applied and computational harmonic analysis*, 21(1):5–30, 2006.

[3] A. W. Long and A. L. Ferguson. Landmark diffusion maps (l-dmaps): Accelerated manifold learning out-of-sample extension. *Applied and Computational Harmonic Analysis*, 47(1):190–211, 2019.

Appendices

A First appendix

B Second appendix